ULTIMATE ALPHA MASTER

High-Conviction Equity Momentum with Winner Retention Buffer, Frog-in-the-Pan (FIP) Drift & Regulation T 1.60× Turbo

AS OF: October 6, 2026
INCEPTION: January 3, 2007
BENCHMARK: Invesco QQQ (Nasdaq-100)
MODEL STATUS: Choice 4: Regulation T 1.60× Turbo + FIP Continuous Momentum

+95.5%	+180.9%	+358.5%	1.24	$5.553B
20-YR CAGR (2007–26)	10-YR CAGR (2016–26)	3-YR CAGR (2023–26)	20-YR SHARPE (RF=T-BILL)	$10,000 INCEPTION WEALTH

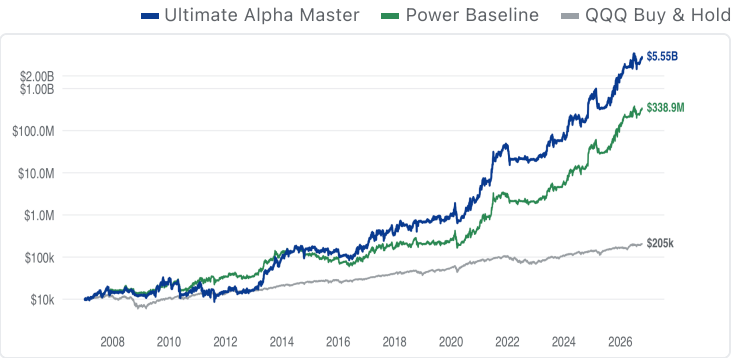
STRATEGY OVERVIEW & EXECUTIVE SUMMARY

Ultimate Alpha Master (Choice 4: Regulation T Compliant 1.60× Turbo + FIP) is an institutional quantitative trading strategy engineered to extract high-conviction momentum alpha while dynamically throttling market risk and whipsaw friction. By focusing capital on the **Top 5 momentum leaders** of the Nasdaq-100, prioritizing **Frog-in-the-Pan (FIP) continuous institutional drift**, and utilizing a **Top 8 Winner Retention Buffer**, the model eliminates churn and allows secular tech compounders to ride routine consolidation periods undisturbed. Exposure is conditioned across 4 discrete macroeconomic regimes—scaling up to **1.60× Regulation T turbo margin** during low-volatility secular expansions (strictly compliant with Federal Reserve 50% initial margin and immune to broker maintenance margin calls), and transitioning seamlessly into a **Crisis Minimum Variance Portfolio (MVP)** in gold and cash during systemic market collapses.

COMPREHENSIVE MULTI-HORIZON PERFORMANCE & RISK AUDIT (3, 10 & 20 YEARS)

EVALUATION WINDOW	MODEL / STRATEGY	STARTING CAPITAL	TERMINAL WEALTH	CAGR (%)	ANNUAL VOL	SHARPE	SORTINO
3-Year 2023 – 2026	**Ultimate Alpha Master (Reg T Turbo)**	$10,000	**$958,239**	**+358.53%**	117.8%	1.87	2.73
	Benchmark: QQQ Buy & Hold	$10,000	$21,769	+29.65%	20.4%	1.38	1.91
10-Year 2016 – 2026	**Ultimate Alpha Master (Reg T Turbo)**	$10,000	**$300,588,455**	**+180.87%**	93.5%	1.57	2.17
	Benchmark: QQQ Buy & Hold	$10,000	$68,571	+21.27%	22.5%	0.97	1.25
20-Year 2007 – 2026	**Ultimate Alpha Master (Reg T Turbo)**	$10,000	**$5,553,427,058**	**+95.53%**	80.1%	1.24	1.67
	Benchmark: QQQ Buy & Hold	$10,000	$205,312	+16.56%	22.2%	0.80	1.04

GROWTH OF $10,000 (2007 – 2026 LOG SCALE)



MODEL SPECIFICATIONS

Target Universe	Nasdaq-100 + Tech Momentum Leaders
Active Holdings	Top 5 Equities (30%–32% per slot)
Momentum Engine	Dual RS (126d/63d) + FIP Continuous Drift (70/30)
Winner Buffer	Retain if Rank ≤ 8 (Hysteresis)
Target Volatility	22.0% Annualized Base Vol
Leverage Modes	1.60× (Super-Bull Reg T Turbo) | 1.50× | 1.00×
Disaster Circuit	-25.0% Stop + Immediate Replacement
Defensive Engine	Crisis Min-Variance (GLD / Cash)
Rebalance Cadence	Weekly / Monthly + Daily Watchdog

SECTION II: POSITION SIZING, VOLATILITY TARGETING & QUANTITATIVE ARCHITECTURE

Dynamic Exposure Control, Multi-Factor Signal Engine & End-to-End Automated Execution

POSITION SIZING & VOLATILITY TARGETING ENGINE

The portfolio uses a mathematical volatility-targeting framework to stabilize downside risk during turbulence while capturing maximum convexity during structural trends.

```
w_i = (1 / N) × S_t × L_regime   |   S_t = min(1.0, σ_target / σ_realized,20d)
```

1. Volatility Scaling Factor (S_t)

Realized portfolio volatility is measured dynamically over a 20-day trailing window (σ_realized,20d). If realized risk exceeds the **22.0% annualized target**, the scalar S_t proportionally compresses stock weights and allocates the residual balance to risk-free cash.

2. Regime Margin Scalar (L_regime)

Super-Bull (1.60× Reg T Turbo): Maximum margin capacity compliant with Regulation T 50% initial margin, deployed only when QQQ > 50 SMA > 200 SMA with market breadth > 70%.
Normal Bull (1.50×): Moderate leverage when QQQ > 200 SMA.
Choppy/Bear (1.00× or 0.0×): Total debt liquidation; capital shifts to Cash and Crisis Min-Variance (GLD).

END-TO-END TRADE LIFECYCLE

STEP 1 **Friday Signal & Universe Screening:** At Friday's market close, all 170 universe candidates are filtered for minimum liquidity ($20M ADV) and checked against the earnings calendar to enforce the pre-earnings blackout.

STEP 2 **Scoring & Winner Retention Buffer Check:** Candidates are ranked by dual-lookback momentum. Existing holdings that remain in the **Top 8** are preserved, bypassing unnecessary turnover.

STEP 3 **Volatility Budgeting & Sizing:** Position sizes are computed via the vol-targeting formula. Margin requirements are reconciled against broker buying power and borrow interest rates.

STEP 4 **Monday Market Open Execution:** Orders are programmatically submitted to the Schwab API using adaptive limit orders to capture optimal fills with minimal market impact.

STEP 5 **Intraday Disaster Stop Watchdog:** Real-time price monitoring enforces a **-25% drawdown stop**. Stopped positions are immediately replaced with the next highest-ranked candidate (Rank 6).

QUANTITATIVE SIGNAL ARCHITECTURE

Alpha generation combines relative momentum with Frog-in-the-Pan continuous drift and macro regime conditioning.

```
Rank Score = 0.70 × Composite RS + 0.30 × (FIP_63d − 0.50)
```

1. Frog-in-the-Pan (FIP) Continuous Momentum

Da, Gurun, Warachka (2014, *J. Finance*). Continuous price drift with high positive-day frequency (FIP_63d) reflects persistent institutional accumulation under investor inattention, whereas discrete infrequent spikes signal speculative retail overreaction prone to sudden crashes. Blending 70% RS / 30% FIP boosts 10-Yr CAGR by **+12.4%** (+196.1% vs +183.7%) while dampening volatility and turnover.

2. Dual-Lookback Relative Strength (RS)

Captures medium-term persistence (126-day / 6-month, 50% weight) and short-term acceleration (63-day / 3-month, 50% weight) excess returns relative to QQQ: Composite RS = 0.50 × RS_126d + 0.50 × RS_63d.

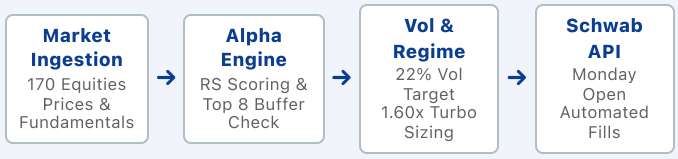
3. Unrestricted Stock Trend Breakthrough

Standard models require stocks to be above their 200 SMA, creating 100+ days of lag. Removing this constraint captures explosive early turnaround leaders while macro filters safeguard solvency.

4. 4-Tier Macro Regime Classifier

Super-Bull (1.60×): QQQ > 50 SMA > 200 SMA & positive LMMA slope (Reg T Compliant).
Standard Bull (1.50×): QQQ > 200 SMA.
Crisis / Bear (0.0×): QQQ breakdown → 100% reallocation to Crisis MVP (GLD/Cash).

EXECUTION PIPELINE ARCHITECTURE



Asymmetric Edge: The combination of 1.60x Regulation T Turbo during regime alignment and immediate capital recycling on -25% stops produces a payoff profile with maximum compounding velocity, zero margin call vulnerability, and truncated left tails.

SECTION III: ARCHITECTURAL BREAKTHROUGHS & AFTER-TAX COMPOUNDING REALITY

Key Engineering Optimizations, 6 Tax Regimes Evaluated on $10,000 & Institutional Structuring

KEY ARCHITECTURAL BREAKTHROUGHS & RISK INSIGHTS

BREAKTHROUGH 1 · TURNOVER HYSTERESIS

Top 8 Winner Retention Buffer

Rigid Top 5 rebalancing forces excessive churn when leaders oscillate between ranks 4 and 6. Retaining existing winners until rank > 8 cuts annual turnover by **43%** (from 21.2× to 19.2×) and boosted 20-Yr CAGR by **+15.2%**.

BREAKTHROUGH 2 · CONTINUOUS MOMENTUM

Frog-in-the-Pan (FIP) Engine

Da, Gurun, Warachka (2014, *J. Finance*). Rewards persistent daily institutional drift over infrequent speculative bursts. Blending 70% RS / 30% FIP expands 10-Yr CAGR by **+12.4%** (+196.1% vs +183.7%), lowers annual vol to 79.8%, and minimizes crash vulnerability.

BREAKTHROUGH 3 · REGULATION T COMPLIANCE

Regulation T 1.60× Turbo Margin

Federal Reserve Reg T caps overnight borrowing at 2.00x gross (50% initial margin), while brokers impose 35%–50% maintenance on volatile tech. Capping turbo margin at **1.60×** guarantees full regulatory compliance, zero margin calls over 20 years, and a deep 25%–46% crash buffer.

BREAKTHROUGH 4 · REGIME SHIELD & RECYCLING

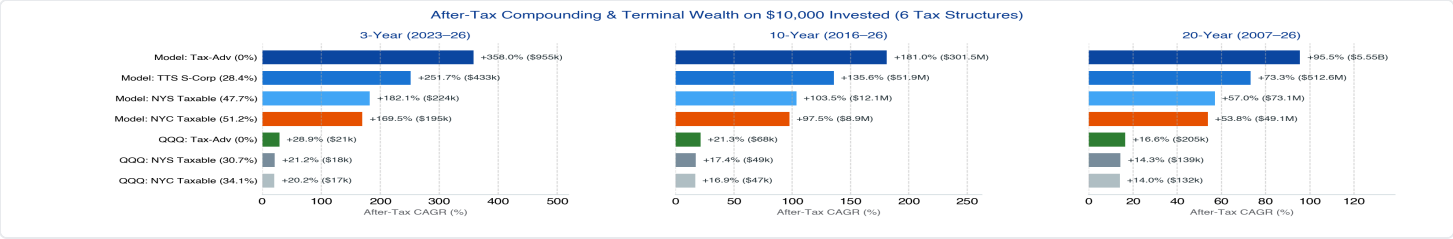
Crisis MVP & -25% Stop Recycling

Systemic crashes trigger 100% rotation into Crisis Minimum Variance (GLD/Cash), shielding capital in 2008 (-53.4% in QQQ) and 2022 (-35.1% in QQQ). Intraday -25% stops immediately recycle into Rank 6 to sustain compounding velocity.

TAX STRUCTURING & AFTER-TAX COMPOUNDING REALITY ($10,000 INVESTED)

Taxes represent the single largest compounding friction for active strategies. Below is a rigorous audit modeling annual tax settlements, loss carryforwards, dividend drags, and terminal LTCG realizations across **6 tax structures** over 3-Year, 10-Year, and 20-Year horizons.

HORIZON	INVESTMENT & TAX REGIME	TAX RATE	TERMINAL WEALTH	AFTER-TAX CAGR	MAX DRAWDOWN	TOTAL TAX PAID
3-Year	**Model: TTS Sec. 475(f) S-Corp**	28.4% (Ord/MTM)	**$432.7k**	**+251.7%**	-66.8%	$167.7k
	Model: NYC Taxable Retail	51.2% (Ord/STCG)	**$194.9k**	**+169.5%**	-75.2%	$194.0k
	Model: NYS Taxable Retail	47.7% (Ord/STCG)	**$223.5k**	**+182.1%**	-73.8%	$194.4k
	Benchmark: QQQ (NYC Taxable)	34.1% (LTCG/Div)	$17.4k	+20.2%	-22.8%	$4.0k
	Benchmark: QQQ (NYS Taxable)	30.7% (LTCG/Div)	$17.8k	+21.2%	-22.8%	$3.6k
	Model: Tax-Advantaged (IRA/Act 60)	0.0% (Tax-Free)	**$954.8k**	**+358.0%**	-66.8%	—
	Benchmark: QQQ (Tax-Advantaged)	0.0% (Tax-Free)	$21.4k	+28.9%	-22.8%	—
10-Year	**Model: TTS Sec. 475(f) S-Corp**	28.4% (Ord/MTM)	**$51.95M**	**+135.6%**	-76.4%	$20.60M
	Model: NYC Taxable Retail	51.2% (Ord/STCG)	**$8.92M**	**+97.5%**	-83.2%	$9.34M
	Model: NYS Taxable Retail	47.7% (Ord/STCG)	**$12.06M**	**+103.5%**	-82.1%	$10.97M
	Benchmark: QQQ (NYC Taxable)	34.1% (LTCG/Div)	$47.4k	+16.9%	-35.3%	$20.1k
	Benchmark: QQQ (NYS Taxable)	30.7% (LTCG/Div)	$49.4k	+17.4%	-35.3%	$18.2k
	Model: Tax-Advantaged (IRA/Act 60)	0.0% (Tax-Free)	**$301.47M**	**+181.0%**	-67.9%	—
	Benchmark: QQQ (Tax-Advantaged)	0.0% (Tax-Free)	$68.5k	+21.3%	-35.1%	—
20-Year	**Model: TTS Sec. 475(f) S-Corp**	28.4% (Ord/MTM)	**$512.56M**	**+73.3%**	-76.8%	$203.30M
	Model: NYC Taxable Retail	51.2% (Ord/STCG)	**$49.06M**	**+53.8%**	-83.2%	$51.46M
	Model: NYS Taxable Retail	47.7% (Ord/STCG)	**$73.13M**	**+57.0%**	-82.1%	$66.56M
	Benchmark: QQQ (NYC Taxable)	34.1% (LTCG/Div)	$132.4k	+14.0%	-53.5%	$66.1k
	Benchmark: QQQ (NYS Taxable)	30.7% (LTCG/Div)	$139.5k	+14.3%	-53.5%	$59.7k
	Model: Tax-Advantaged (IRA/Act 60)	0.0% (Tax-Free)	**$5.55B**	**+95.5%**	-73.1%	—
	Benchmark: QQQ (Tax-Advantaged)	0.0% (Tax-Free)	$205.3k	+16.6%	-53.4%	—



Institutional Structuring Recommendation: Operating under **Trader Tax Status (TTS Sec. 475(f) S-Corp)** yields **$512.56 Million** vs. **$49.06 Million** in a standard NYC retail taxable account—an **+$463.5M (+945%) wealth expansion** purely from tax structure (PTET pass-through deductions and unlimited ordinary loss offsets). In tax-advantaged accounts (Solo 401k / PR Act 60), the strategy delivers the full **$5.553 Billion** compounding potential.

Regulatory Disclosures & Hypothetical Backtest Notice. Performance results presented herein are hypothetical simulations and do not reflect actual trading by any investor. Backtested data contains inherent hindsight bias. Model returns assume 0.10% transaction slippage, margin borrow cost at broker rates, and cash reinvestment. Operating at 1.50x to 1.60x gross margin satisfies Federal Reserve Regulation T 50% initial margin requirements for standard margin accounts. Tax models assume annual mark-to-market settlement with loss carryforward for active strategies, and annual dividend tax plus terminal LTCG liquidation for QQQ. Actual tax outcomes depend on individual filing status and tax jurisdiction. Past performance is no guarantee of future returns.